



Special issue in honor of Prof. Suleyman I. Allakhverdiev

REVIEW

Salt stress in cyanobacteria and its relevance to energy management

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Abstract

Energy management in photosynthetic organisms occurs at multiple levels, including the electron transfer and metabolic reactions. Salt stress imposes osmotic, ionic, energetic, and oxidative constraints on cyanobacteria, profoundly reshaping carbon allocation and photosynthetic performance. To maintain osmotic balance, cyanobacteria employ a salt-out strategy, accumulating compatible solutes such as glucosylglycerol and sucrose while maintaining low intracellular Na⁺ concentrations. This review provides a comprehensive perspective by synthesizing current knowledge on cyanobacterial overflow under salt stress, emphasizing its physiological context rather than focusing on a single dedicated pathway. We compare metabolite overflow responses under nitrogen limitation, high light, mixotrophy, and combined stresses, and discuss methodological approaches integrating targeted analytics, untargeted exometabolomics, energy charge measurements, and flux analysis. Collectively, energy management *via* metabolite overflow under salt stress is proposed as an emergent, condition-dependent outcome of integrated carbon, energy, and redox regulation.

Keywords: carbon imbalance; cyanobacteria; energy management; metabolism; metabolite overflow; salinity.

Introduction

Cyanobacteria, algae, and green plants can harvest sunlight and store it as chemical energy. Specifically, they capture sunlight energy and transfer it to the energy carriers ATP

and NADPH in living cells (Long *et al.* 2015, Ort *et al.* 2015, Hou *et al.* 2017, Nocera 2017, Shen *et al.* 2021, Hou and Allakhverdiev 2023). Photosynthetic energy is regulated and managed *via* a variety of mechanisms to maintain normal growth and to survive environmental

Highlights

- Salt stress imposes energetic constraints and reshapes photosynthetic performance
- Energy management occurs at the levels of electron transfer and metabolism
- Response to salt stress is proposed for biomanufacturing *via* energy management

Received 30 March 2026

Accepted 11 June 2026

Published online 30 July 2026

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Abbreviations: 2-OG – 2-oxoglutarate; ATP – adenosine triphosphate; GC–MS – gas chromatography–mass spectrometry; HPLC – high-performance liquid chromatography; LC–MS/MS – liquid chromatography–tandem mass spectrometry; MFA – metabolic flux analysis; PHB – polyhydroxybutyrate; PII – nitrogen regulatory protein PII; ROS – reactive oxygen species; TCA – tricarboxylic acid.

Acknowledgments: We thank Barry Bruce and two anonymous reviewers for their constructive critiques to improve the quality of the manuscript. We appreciate Himadri Pakrasi, Fuzhong Zhang, Yinjie Tang, Yixin Chen, Katherine Chou, and B.K. Robertson *via* the tri-institutional (NLR–WashU–ASU) cyanobacteria team for stimulating discussion. This work was supported by the U.S. Department of Energy, SFEE program (DE-SC002567) and RENEW program (DE-SC0024702), and in part by the National Science Foundation (NSF MCB-2428256). This work was authored in part by the National Laboratory of the Rockies for the U.S. Department of Energy (DOE) under Contract No. DE-AC36-08GO28308. Funding provided by the U.S. Department of Energy Office of Science Office of Biological and Environmental Science under Secure Biosystems Design Science Focus Area (SFA). The views expressed in the article do not necessarily represent the views of the DOE or the U.S. Government. The U.S. Government retains, and the publisher, by accepting the article for publication, acknowledges that the U.S. Government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this work, or allow others to do so, for U.S. Government purposes.

Conflict of interest: The authors declare that they have no conflict of interest.

stress (Aro *et al.* 1993, Baker and Bowyer 1994). Energy management occurs at multiple levels, including at the level of electron transfer and at the level of metabolic reactions. Mechanisms at the electron transfer level have been investigated for decades, including the xanthophyll cycle, nonphotochemical quenching, cyclic and alternative electron transfer, *etc.* (Aro *et al.* 1993, Baker and Bowyer 1994, Long *et al.* 1994, Niyogi 1999, Kramer *et al.* 2003). In contrast, energy management mechanisms at the metabolic level are less understood. Carbon fixation and metabolism also play an important role in energy regulation.

As biomass growth is often limited by nutrient limitation or other environmental conditions, a significant amount of organic compounds can be secreted into the medium in what is known as overflow metabolism. We have reported the overflow metabolism in mutant strains of cyanobacteria to study its potential in producing organic acids, and to identify such a metabolism as an energy regulation mechanism (Carrieri *et al.* 2012, Cano *et al.* 2018).

Most living cells can only survive within a limited range of salinity. While life is believed to have originated in the saline environment of early oceans, modern organisms exhibit significant differences in their tolerance to salt (Oren 2014, Gunarathne *et al.* 2023). Salinity remains a crucial environmental factor shaping aquatic ecosystems and is increasingly a serious constraint in terrestrial environments due to the global expansion of soil salinization (Gunarathne *et al.* 2023). As a result, salt acclimation has garnered long-standing interest not only in ecology but also in biotechnology, particularly for improving salt tolerance in crop plants (Erdmann and Hagemann 2001).

Algae and cyanobacteria are central to salt-acclimation research. These photosynthetic microorganisms naturally inhabit environments with high, variable, or compositionally complex salinity, ranging from freshwater to marine and hypersaline systems. In addition to their ecological relevance, cyanobacteria serve as useful model organisms for understanding how photosynthetic cells respond to osmotic and ionic stress. This is due to their relatively simple cellular organization, short generation times, and well-developed genetic tools (Santos-Merino *et al.* 2021). Many characteristics of cyanobacterial photosynthesis, membrane composition, and signal transduction closely resemble those of chloroplasts in higher plants, making them valuable models for plant stress biology (Gross *et al.* 2012).

In cyanobacteria, an increase in external salinity poses two immediate and interconnected challenges. First, reduced water availability leads to water loss from the cell, resulting in a decreased cell volume and turgor pressure. Second, elevated concentrations of inorganic ions increase the passive influx of ions such as Na⁺ and Cl⁻, disrupting enzyme activity, membrane potential, and ionic homeostasis (Yang *et al.* 2020). Successful acclimation, therefore, requires a coordinated cellular response aimed at restoring osmotic balance and cell volume, re-establishing ion homeostasis through regulated transport and membrane

adjustments, and protecting proteins, nucleic acids, and membranes from the combined effects of high ionic strength and reduced water activity (Klähn and Hagemann 2011). In most cyanobacteria, these goals are achieved through a salt-out strategy, in which intracellular inorganic ion concentrations are kept low while osmotic adjustment is facilitated by the accumulation of compatible organic solutes. For example, in *Synechocystis* sp. PCC 6803, glucosylglycerol, and sucrose serve as major osmolytes (Hagemann 2011). The transcriptomics analysis revealed that the salt stress response is energetically costly and involves extensive changes in gene expression, protein turnover, antioxidant defenses, and central metabolism (Hagemann 2011, Klähn and Hagemann 2011).

Salt stress is also closely linked to photosynthetic function. Elevated Na⁺ and Cl⁻ concentrations impair electron transport, disrupt cellular K⁺/Na⁺ balance, and promote the formation of reactive oxygen species (ROS), ultimately damaging the photosynthetic apparatus (Long *et al.* 1994). The short-term salt stress is an acute shock response occurring within minutes to hours, and the long-term salt acclimation is a steady-state physiological adaptation over extended periods. This distinction is critical, as metabolic and photosynthetic responses differ substantially between these two states. For example, under short-term salt stress, D1 protein repair inhibition contributes to photoinhibition; however, in long-term salt-acclimated cells, D1 turnover rates can recover to levels comparable to non-stressed conditions. Since cyanobacteria depend on a continuous light-driven electron flow, this creates a fundamental physiological conflict: under salt stress, cells must allocate more energy towards homeostasis and protection, while the photosynthetic machinery becomes increasingly vulnerable to over-reduction and oxidative damage (Long *et al.* 1994). Understanding how cyanobacteria resolve this imbalance is a central question in stress physiology.

One adaptive response that has attracted attention is metabolite overflow, defined as the excretion of fixed carbon as small organic compounds when photosynthetic carbon input exceeds the cell's capacity for growth or stable storage (Hudson 2021). Rather than representing a simple loss of carbon, overflow metabolism is now considered a regulated strategy that stabilizes cellular redox state and energy balance under metabolic constraints. Disruption of major carbon storage sinks, such as glycogen, leads to the extracellular accumulation of metabolites, including pyruvate and 2-oxoglutarate. Similar excretion patterns can occur due to high light or nutrient imbalances (Cano *et al.* 2018). Salt stress can promote overflow through several mechanisms: increased ATP demand for ion transport and compatible solute synthesis, altered carbon partitioning away from biomass, and heightened oxidative pressure on the photosynthetic electron transport chain (Allakhverdiev and Murata 2008). At the same time, the biosynthesis of osmolytes can compete for carbon internally, which means salt stress may either suppress or enhance overflow, depending on the species, genetic background, nutrient availability, and light conditions. These opposing influences make salt-associated metabolite

overflow a complex and still incompletely understood phenomenon.

Beyond its physiological significance, metabolite overflow under salt stress has important biotechnological implications. Large-scale cyanobacterial cultivation is often envisioned in seawater-based systems to conserve freshwater resources and reduce contamination risks (Bolay *et al.* 2025). However, salt acclimation imposes substantial energetic costs and can divert carbon away from productivity, while overflow represents both a challenge, due to carbon loss, and an opportunity, as naturally secreted metabolites could be redirected toward valuable products such as organic acids, alcohols, or polyhydroxybutyrate (PHB)/3-hydroxybutyrate-related compounds (Bai *et al.* 2025). Achieving predictable and efficient production under saline conditions, therefore, requires a mechanistic understanding of how osmotic and ionic stress intersect with photosynthetic performance, carbon storage, and metabolite export. In addition, hypoosmotic shock reduction in salinity can activate mechanosensitive channels, resulting in the release of intracellular compatible solutes, a process known as ‘bacterial milking’, which has been explored for metabolite harvesting.

In this review, we provide perspective by synthesizing current knowledge on cyanobacterial energy management under salt stress and integrating physiological, methodological, and mechanistic perspectives. We first summarize the core processes underlying salt acclimation in cyanobacteria, with emphasis on osmotic adjustment, ion transport, and photosynthetic vulnerability. We then describe experimental strategies for detecting and quantifying overflow, including targeted analytical approaches and exometabolomics compatible with high-salinity systems. Next, we compare overflow responses across environmental conditions and genetic backgrounds to identify recurring patterns associated with salinity stress.

By integrating these perspectives, this review aims to provide a framework for interpreting overflow metabolism in natural saline environments and for guiding the rational design of cyanobacterial strains and cultivation strategies for seawater-based bioprocesses.

Physiology of cyanobacteria under salt stress

Cyanobacteria encounter a multifaceted physiological challenge in response to salt stress, which involves complex interactions among osmotic stress, ionic imbalance, and secondary oxidative and metabolic stresses rather than a singular disturbance (Erdmann and Hagemann 2001, Yang *et al.* 2020). An increase in external salinity exposes these microorganisms to a variety of physiological responses; namely, the influx of sodium (Na^+) and chloride (Cl^-) ions occurs simultaneously (Fig. 1). These stresses are not independent; rather, they interact in intricate ways that influence both the immediate cellular responses to elevated salinity and the long-term acclimatization processes that enable growth to resume in saline environments. While most of our in-depth knowledge arises from a relatively small number of extensively studied model strains, the fundamental physiological strategies employed to cope with salt stress are notably conserved across diverse cyanobacterial species inhabiting freshwater, brackish, marine, and hypersaline ecosystems.

Early salt shock responses: rapid osmotic and ionic imbalance: When cyanobacteria are abruptly exposed to elevated salinity, they undergo an immediate phase of osmotic shock, marked by rapid water efflux from the cell due to a steep osmotic gradient (Fig. 1). This loss results in a significant reduction in cell volume and turgor pressure. If this loss of turgor is not promptly compensated, it can destabilize membranes and compromise the structural

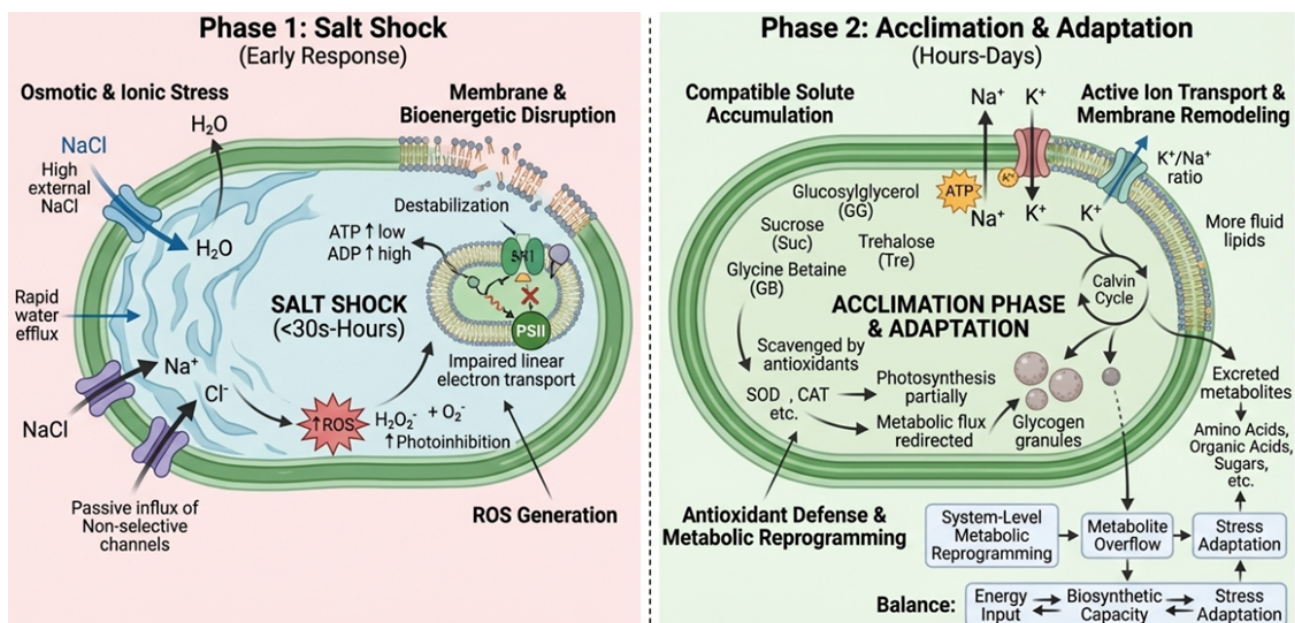


Fig. 1. Two phases of response mechanisms of cyanobacteria under salt stress conditions.

integrity of the cellular architecture (Pade and Hagemann 2015). Additionally, the elevated concentrations of Na^+ and Cl^- ions in the external environment facilitate their passive influx into the cytoplasm, driven by established electrochemical gradients. The uncontrolled accumulation of these ions poses a substantial threat to cellular bioenergetics and viability by disrupting enzyme activities, membrane potential, and intracellular pH. This acute phase of the salt shock occurs within seconds to minutes and is physiologically distinct from the subsequent acclimatization period. Transcriptomic and proteomic studies consistently indicate that numerous stress-responsive genes and proteins may be transiently induced during this initial response phase, signifying an emergency mobilization of cellular resources rather than a transition toward stable metabolic adaptation (Yu *et al.* 2024). As the duration of salt exposure increases, many of these early physiological alterations diminish, suggesting that measurements taken shortly after the onset of salt exposure may not accurately reflect the physiology of cells that have transitioned to acclimation.

Osmotic adjustment via compatible solutes: To restore turgor pressure while avoiding toxic intracellular ion concentrations, most cyanobacteria employ a salt-out strategy. To avoid high internal concentration of inorganic ions, Na^+ is actively exported, while osmotic adjustment is achieved by synthesizing or importing compatible solutes, small organic molecules that can accumulate to high concentrations without impairing enzymatic function. The specific types of compatible solutes produced generally correspond to the level of salt tolerance in that species. For example, many moderately halotolerant species predominantly synthesize glucosylglycerol, while species with low salt tolerance use trehalose or sucrose, and halophiles use glycine betaine or glutamate (Pade *et al.* 2015). The metabolic investment required for the synthesis of these compatible solutes is substantial, necessitating significant amounts of fixed carbon, adenosine triphosphate (ATP), and reducing power. This leads to a pronounced reorganization of central carbon metabolism during the acclimatization process (Furmaniak *et al.* 2017, Klähn *et al.* 2021). The diversion of metabolic resources toward osmolyte production disrupts the typical balance between growth, storage, and other metabolic obligations. Consequently, cyanobacteria that have acclimated to salt stress frequently exhibit reduced growth rates and distinct intracellular and extracellular metabolite profiles compared to unstressed cells, even after restoration of osmotic balance.

Ion homeostasis and membrane transport: The accumulation of compatible solutes is insufficient for survival in saline conditions due to the ongoing influx of Na^+ and Cl^- ions. Thus, it becomes imperative for cyanobacteria to actively regulate ion transport to prevent cytotoxic concentrations of these ions from accumulating within the cell. Maintaining a high intracellular potassium (K^+)/sodium (Na^+) ratio is especially critical, as numerous enzymes, ribosomes, and pivotal metabolic processes

are exceedingly sensitive to elevated Na^+ concentrations (Roychoudhury *et al.* 2025). Ion homeostasis is achieved through the coordinated activity of various membrane transport systems, including Na^+/H^+ antiporters and other specific ion transporters. The energy demands of these systems are considerable, as the export of ions is coupled to proton gradients generated by both photosynthetic and respiratory electron transport processes. The maintenance energy cost of salt stress is $10\times$ the energy cost without salt stress in *E. coli* (Schink *et al.* 2024). Furthermore, salt stress alters membrane lipid composition and transporter abundance, thereby influencing membrane permeability and transport efficiency (Roychoudhury *et al.* 2025). These findings underscore the notion that membranes serve not only as passive barriers but as integral regulatory platforms essential for acclimation to marine environments.

Effects of salt stress on membranes and the photosynthetic apparatus: The plasma membrane and thylakoid membrane system are directly affected by salt stress (Sudhir *et al.* 2005). Changes in ionic strength and water activity impact membrane fluidity, protein–lipid interactions, and the stability of large photosynthetic complexes. Because cyanobacteria carry out both photosynthesis and a significant portion of respiratory electron transport within the thylakoid membrane, perturbations at this level have system-wide consequences (Agrawal *et al.* 2015). Experimental studies show that salt stress disrupts electron transport in both Photosystem II (PSII) and Photosystem I (PSI). Altered K^+/Na^+ ratios and direct ionic effects on protein complexes impair electron flow, increasing susceptibility to photoinhibition. Importantly, salt stress does not primarily accelerate photodamage itself but instead inhibits the repair of damaged PSII by suppressing the synthesis of key components, particularly the D1 protein encoded by the *psbA* genes (Allakhverdiev *et al.* 2002, Marin *et al.* 2003, Allakhverdiev and Murata 2008). As a result, photoinhibition becomes more severe under illumination, even at moderate salt concentrations. These responses are not unique to salt stress but are part of a broader general stress response shared across multiple abiotic stress conditions.

Oxidative stress as a secondary consequence: The disruption of photosynthetic electron transport caused by salt stress significantly impacts cyanobacterial physiology by leading to an over-reduction of the electron transport chain. This over-reduction facilitates the increased production of reactive oxygen species (ROS), primarily at the PSII and PSI complexes. The phenomenon is particularly pronounced when light absorption persists at high levels despite reduced metabolic demand, leading to unchecked ROS accumulation that can adversely affect cellular integrity and function (Liu *et al.* 2021). Elevated concentrations of ROS are highly damaging; they can cause oxidative damage to vital cellular structures, such as proteins, lipids, and nucleic acids. This oxidative damage further undermines photosynthetic efficiency and disrupts membrane integrity, thereby harming

cellular health. Moreover, high ROS concentrations can inhibit protein synthesis by targeting essential translation factors, thereby exacerbating the inhibition of PSII repair mechanisms and intensifying photoinhibition, a state in which the photosynthetic apparatus is subjected to stress and impaired in the light-energized environment (Foyer 2018). To combat the harmful effects of oxidative damage, cyanobacteria have evolved a range of protective strategies. These mechanisms encompass the activation of various antioxidant enzymes that scavenge ROS, the accumulation of carotenoids that provide photoprotection, the activation of energy-dissipating pathways that help regulate excess excitation energy, and the synthesis of photoprotective proteins, all of which work synergistically to mitigate oxidative stress.

Two-phase acclimation and system-level reprogramming: Cyanobacterial acclimation to salt stress can be described in terms of two distinct but interrelated phases (Fig. 1). The initial shock phase is marked by rapid physiological responses, characterized by extensive alterations in gene expression and metabolic pathways that protect the organism from immediate cellular damage. During this phase, there is an influx of signaling that triggers an array of protective measures aimed at stabilizing cellular function under stress. Following this initial response, the organism enters a longer-term adaptation phase, during which a narrower and more specialized suite of processes is sustained. This includes the synthesis of compatible solutes that help maintain cellular osmotic balance, the activation of ion transport mechanisms to manage toxic ion concentrations, membrane remodeling to enhance cellular resilience, and the fortification of antioxidant defenses to counteract persistent oxidative stress (Hagemann 2011, Klähn and Hagemann 2011, Kim *et al.* 2016, Yang *et al.* 2020). Comprehensive system-level analyses have revealed that acclimating to saline environments necessitates a global reallocation of cellular resources. In response to increasing energy demands for ion pumping and repair processes, carbon flux is redirected away from biomass production toward maintenance and protective functions. This strategic reallocation implies that photosynthetic output must be meticulously controlled to prevent a redox imbalance that could compromise cellular integrity. Thus, the cell's metabolic landscape undergoes significant restructuring under saline conditions, reflecting the organism's acute adaptability to environmental stressors (Erdmann and Hagemann 2001, Hagemann 2011, Klähn and Hagemann 2011, Yang *et al.* 2020).

Physiological context for energy management under salt stress: From a physiological standpoint, salt stress engenders a complex interplay between energy availability and biosynthetic capacity that is often untenable. The photosystems in cyanobacteria may continue to capture light energy effectively; however, growth and macromolecular synthesis are constrained by osmotic, ionic, and oxidative challenges. Under these adverse conditions, the synthesis of compatible solutes, which serve as protective osmotic agents, starts to compete

with other carbon sinks within the cell. Concurrently, the associated maintenance energy costs rise sharply, leading to a resource balance precariously adjusted. The overflow of pyruvate and 2-oxoglutarate observed in glycogen mutants during N-starvation is abolished by adding salt stress because the synthesis of glucosylglycerol replaces the sink (Díaz-Troya *et al.* 2020).

In response to the physiological challenges posed by salt stress, cyanobacteria may redirect excess fixed carbon into storage polymers, a strategy that helps preserve some nutrients. In certain instances, they may also excrete low-molecular-mass metabolites into the surrounding environment to manage internal stress. The extent to which salt stress facilitates or inhibits this metabolite overflow depends on a delicate balance among osmolyte production capacity, available storage capacity, and overall photosynthetic activity. It is crucial to understand that overflow metabolism under salt stress does not arise as a straightforward or inevitable result of increased salinity; rather, it is an emergent property that reflects the intricate, integrated nature of cellular physiology under challenging environmental conditions. To date, there is limited direct evidence demonstrating metabolite excretion under steady-state salt-acclimated conditions (Carrieri *et al.* 2010, Díaz-Troya *et al.* 2020); therefore, this option should be considered a hypothesis requiring further investigation.

Mechanisms of cyanobacterial energy management

The phenomenon of energy management, including metabolite overflow, in cyanobacteria can best be interpreted as a complex interplay of physiological, metabolic, and regulatory mechanisms, rather than a simplistic linear pathway (Chi *et al.* 2019). As shown in Fig. 2, current research suggests that the mechanisms facilitating overflow can be categorized into three conceptual layers: (a) the driving forces that promote overflow, (b) the metabolic branch points at which excess carbon is generated, and (c) the pathways that enable metabolites to be extruded into the extracellular space.

Driving forces

Sink limitation and constrained anabolic capacity: When cyanobacteria are subjected to salt stress, they

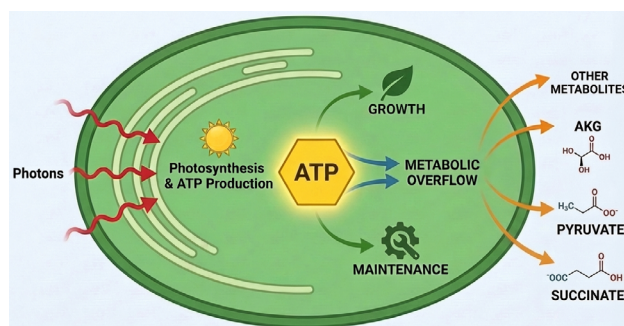


Fig. 2. Simplified scheme of metabolite overflow in cyanobacteria and possible role in energy management.

face significant challenges that impose a substantial energetic and biosynthetic burden. Stress prompts a reallocation of resources towards essential functions such as stress defense and cellular maintenance, specifically ion transport mechanisms, membrane repair processes, antioxidant defense systems, and the production of compatible solutes (Soru 2019). As a result of these diverted resources, processes such as protein synthesis, cell growth, and cellular division commonly slow down. Interestingly, while light harvesting and carbon fixation can persist, the overall anabolic activity is hindered. Under these conditions, anabolic sinks, such as biomass formation and macromolecular synthesis, become limited, leading to an accumulation of fixed carbon within central metabolic intermediates. This accumulation consequently increases the likelihood that carbon will be diverted from intracellular metabolic pathways, ultimately leading to the export of overflow metabolites from the cell (Hudson 2021). Alternatively, even salt-stressed cells do not excrete organic compounds under steady-state conditions, because they build up a high intracellular carbon sink, the compatible solute pool; hence, excess organic carbon should be used to maintain this pool according to the growth rate. While this mechanism is plausible based on current metabolic understanding, direct experimental validation under salt stress conditions remains limited.

Energy and redox balance: Another significant aspect of overflow metabolism is its role in maintaining energy and redox balance amidst metabolic imbalances. The process of exporting reduced carbon compounds can help mitigate elevated levels of intracellular metabolites while also aiding in the dissipation of excess ATP and reduced equivalents generated during photosynthesis (Hudson 2021). This phenomenon has been prominently demonstrated in studies involving glycogen-deficient cyanobacterial mutants, in which the excretion of organic acids is tightly correlated with changes in cellular energy charge, necessitating compensatory adjustments in photosynthetic activity. Under salt stress, this balance can be altered in multiple ways. Increased ATP consumption from processes such as ion pumping and the synthesis of compatible solutes heightens the risk of oxidative stress, especially under illuminated conditions (Jassal *et al.* 2025). Furthermore, the inherent capacity for internal carbon storage sinks might be affected or even temporarily saturated. Collectively, these factors foster an environment in which overflow metabolism emerges as a strategic approach to stabilize cellular energy and maintain redox homeostasis. Consistent with this view, mutant-based studies by Cano *et al.* (2018) that disrupt carbon–nitrogen balance, such as glycogen-deficient strains, demonstrate distinct overflow phenotypes, highlighting the central role of metabolic regulation in controlling metabolite excretion.

Intersection of stress signaling and carbon regulation: The acclimatization of cyanobacteria to high salinity is a highly coordinated process governed by intricate regulatory networks that blend environmental sensing with metabolic control mechanisms. Salt-responsive histidine

kinases, alongside global transcriptional regulators, regulate expression patterns encompassing not only stress defense mechanisms but also key central metabolic pathways (Chaffai *et al.* 2024). Central regulatory systems that interlink carbon and nitrogen metabolism respond to fluctuations in the concentrations of intracellular metabolites such as ATP, ADP, and 2-oxoglutarate (Forchhammer and Selim 2020). This integration of regulatory networks suggests that overflow metabolism may function as an actively modulated response rather than a purely passive outcome of metabolite accumulation. In this context, overflow metabolism reflects an orchestrated redistribution of carbon flux catalyzed by stress signaling networks that prioritize cellular survival at the expense of growth (Hudson 2021). The recent work by Alford *et al.* analyzed overflow metabolisms in different mutants with affected C/N regulation (Alford *et al.* 2025).

Metabolic branch points

Pyruvate as a central metabolic node: Pyruvate plays a pivotal role in the metabolic network of cyanobacteria, serving as a central hub at the convergence of glycolysis, Calvin cycle-derived carbon metabolism, and acetyl-CoA formation. When the pathways responsible for downstream utilization become saturated or limited, pyruvate contents can increase significantly. This excess pyruvate may be excreted directly or rerouted to fermentative or overflow pathways, which may be native to the organism or introduced through metabolic engineering efforts (Pathania and Srivastava 2025). Experimental evidence strongly supports pyruvate as a major branching point for overflow metabolism. For example, Cano *et al.* (2018) demonstrated that a glycogen-deficient *Synechocystis* sp. PCC 6803 mutant (Δ glgC) accumulated high intracellular energy charge and released significant amounts of pyruvate and 2-oxoglutarate into the extracellular medium, particularly under conditions of elevated light input (Foyer 2018). These results indicated that when the primary carbon storage pathway is impaired, central metabolites such as pyruvate become key outlets for excess photosynthetic carbon and energy. Numerous investigations of organic acid production in cyanobacteria have consistently underscored the importance of pyruvate as a critical branching point from which a variety of overflow products can be synthesized (Pathania and Srivastava 2025). In addition to organic acids, several amino acids derived from pyruvate may also contribute to overflow, particularly under nitrogen-limited conditions when the glycogen sink is absent.

Metabolite 2-oxoglutarate as a carbon-nitrogen signal: Metabolite 2-oxoglutarate (2-OG) serves dual functions in cyanobacterial systems, acting as both a crucial metabolic intermediate and a signaling molecule that links carbon status to nitrogen availability. Under nitrogen limitation, 2-OG tends to accumulate and is particularly prone to excretion in strains with impaired carbon storage (Cano *et al.* 2018, Esteves-Ferreira *et al.* 2018). Its intracellular concentrations are finely tuned by nitrogen assimilation

processes and regulatory proteins that detect the carbon-to-nitrogen balance, including pathways governed by PII-dependent signaling networks. The dynamics of 2-OG are likely to be affected by salt stress, even when sufficient external nitrogen is present. Variations in energy status, redox balance, and nitrogen-assimilation efficiency can induce a state resembling functional nitrogen stress, leading to enhanced accumulation of 2-OG and its potential release as an overflow metabolite (Cano *et al.* 2018, Robles-Rengel *et al.* 2019). In the study by Cano *et al.* (2018), extracellular excretion of 2-oxoglutarate and pyruvate occurred not only under nitrogen starvation but also under nitrogen-replete conditions when energy input exceeded cellular assimilation capacity, indicating that energy imbalance alone can trigger overflow metabolism. The experimental evidence for 2-oxoglutarate accumulation is best demonstrated in studies where cells are shifted from nitrate to ammonium conditions.

Carbon storage vs. export: glycogen, PHB, and compatible solutes: Cyanobacteria possess multiple internal carbon sinks that can buffer excess fixed carbon, including glycogen, polyhydroxybutyrate (PHB) in certain species or conditions, and compatible solutes such as glucosylglycerol and sucrose, which are utilized during salt acclimation (Lucius 2024). Among these, glycogen represents a major storage sink that generally suppresses overflow when functioning efficiently. Salt stress increases the demand for compatible solutes, drawing carbon into osmolyte synthesis. A useful working hypothesis is that if the capacity or regulation of compatible solute synthesis is limiting or delayed, particularly during early phases of salt shock, fixed carbon may transiently exceed the buffering capacity of internal sinks and spill into overflow metabolites such as organic acids or other exudates, based on the experimental data from rice (Owen 1987). This hypothesis is not a reliable support for this scenario in salt-shocked cyanobacteria and may be tested using time-resolved exometabolomics combined with measurements of osmolyte pathway enzyme expression and abundance, including key enzymes such as glucosylglycerol-phosphate synthase (Hagemann 2011). This hypothesis remains to be validated specifically in cyanobacteria under salt stress.

Export routes: Overflow metabolism requires not only the production of excess metabolites but also their release into the extracellular environment. Several non-mutually exclusive routes may contribute to this process.

One possibility is transport *via* specific or low-specificity membrane transporters. Some organic acids and sugars are known to be exported by dedicated transport systems, and salt acclimation is accompanied by changes in transporter expression profiles, suggesting a role for regulated export mechanisms. Alternatively, changes in membrane permeability under salt and oxidative stress may facilitate passive leakage. Salt-induced alterations in membrane composition, combined with ROS-mediated damage, can increase membrane permeability and modify efflux behavior, thereby contributing to the accumulation

of extracellular metabolites. A third, species-dependent mechanism involves the release of outer membrane vesicles, which can contribute to extracellular pools of metabolites and small molecules, even if not always classified as classical overflow metabolism. Mechanosensitive channels likely play a critical role in mediating metabolite release during osmotic transitions, particularly during hypoosmotic shock.

A major unresolved challenge in the field is distinguishing between regulated export and damage-associated leakage. Addressing this question requires an integrated experimental approach that combines exometabolomics with viability assays, membrane integrity measurements, and time-resolved stress markers (Pinu *et al.* 2018). Such strategies are essential for determining whether overflow under salt stress represents an adaptive, regulated process or a byproduct of cellular damage or, most likely, a continuum between the two. Cano *et al.* (2018) addressed this question using live/dead cell staining. In that example, live cells were found to overflow metabolites (Cano *et al.* 2018). Additionally, Kirsch *et al.* (2017) reported the role of the glucosylglycerol-degrading enzyme in acclimation to fluctuating salinities.

Overflow metabolism under different conditions

Overflow metabolism in cyanobacteria is not restricted to a single environmental trigger but emerges across a range of conditions that create imbalances among carbon fixation, energy input, and biosynthetic capacity. Studies across model cyanobacteria have shown that nutrient availability, light intensity, carbon source, and salinity all influence whether fixed carbon is retained internally or released into the extracellular environment.

Carbon–nutrient imbalance: nitrogen limitation and glycogen as a primary carbon sink: Nitrogen starvation is one of the best-characterized conditions that promote overflow-like phenotypes in cyanobacteria. Under nitrogen deprivation, photosynthetic carbon fixation can continue for some time, while growth and protein synthesis slow down due to limited nitrogen availability. Many cyanobacteria respond by diverting excess fixed carbon into glycogen, which serves as a primary storage sink and helps stabilize cellular energy balance. When glycogen synthesis is disrupted, however, this buffering capacity is lost, and carbon is forced into alternative pathways, including extracellular excretion.

A landmark study demonstrated that glycogen synthesis and metabolite overflow jointly contribute to energy balancing in *Synechocystis* sp. PCC 6803. Glycogen-deficient strains excreted organic acids, particularly pyruvate and 2-oxoglutarate, under nitrogen starvation and, notably, even under nitrogen-replete conditions when exposed to high light, provided that the energy input exceeded a critical threshold (Gründel *et al.* 2012). Complementary work in *Synechococcus elongatus* PCC 7942 further examined the factors controlling pyruvate excretion in glycogen storage mutants, highlighting that overflow depends strongly on the growth regime,

light intensity, and nutrient status, rather than nitrogen availability alone (Benson *et al.* 2016). More recent studies have continued to refine this picture, emphasizing the central role of glycogen as a metabolic buffer that suppresses overflow under balanced conditions. Together, these findings establish glycogen as a primary carbon sink that limits overflow. When glycogen capacity is reduced or eliminated, overflow metabolism becomes a prominent alternative route for excess fixed carbon.

Recently, Cano *et al.* (2018) suggested that additional carbon sinks can also participate in cellular energy management. Manipulation of glycogen and sucrose synthesis pathways has been shown to influence photosynthetic productivity, indicating that these pathways may function as ATP-consuming cycles that help balance photosynthetic energy input with metabolic demand (Carrieri *et al.* 2012, Cano *et al.* 2018). Reducing flux through glycogen or sucrose cycling can redirect energy toward growth, demonstrating that these storage pathways compete with biomass production for photosynthetically derived energy (Cantrell *et al.* 2023). Energy buffering in cyanobacteria is not limited to carbon metabolism. Polyphosphate metabolism has also been implicated in energy management. Polyphosphate synthesis and degradation can dissipate ATP, providing an additional regulatory layer for maintaining energy homeostasis. Deletion of the *ppk* gene, encoding polyphosphate kinase, has been shown to increase ATP contents and enhance photosynthetic productivity, suggesting that polyphosphate cycling normally acts as an energy-consuming buffer that stabilizes intracellular energy balance (Sebesta *et al.* 2024).

Energy buffering in cyanobacteria is not limited to carbon metabolism. Polyphosphate metabolism has also been implicated in energy management. Polyphosphate synthesis and degradation can dissipate ATP energy, providing an additional regulatory layer for maintaining energy homeostasis. Deletion of the *ppk* gene, encoding polyphosphate kinase, has been shown to increase ATP levels and enhance photosynthetic productivity, suggesting that polyphosphate cycling normally acts as an energy-consuming buffer that helps stabilize intracellular energy balance (Sebesta *et al.* 2024).

High light and energy-driven overflow: High light intensity can also trigger overflow metabolism in the absence of nutrient starvation. Under such conditions, photosynthetic electron flow and ATP generation increase, while downstream anabolic sinks may become saturated. In glycogen-deficient backgrounds, high light alone is sufficient to induce organic acid excretion, supporting the concept of energy overflow driven by excess light input (Zhang *et al.* 2021). These observations suggest a threshold behavior, whereby cyanobacteria export metabolites once internal mechanisms for dissipating excess energy and reducing power are exceeded. Notably, the comprehensive work of the Ducat group demonstrated that engineering sucrose export creates an external carbon sink with multiple consequences, enhancing photosynthetic activities and metabolic output (Santos-Merino *et al.* 2021).

Mixotrophy and external carbon supply: Mixotrophic growth, such as supplementation with glucose or other organic carbon sources, further alters carbon flux and redox balance. Under these conditions, carbon enters the metabolic network through both CO₂ fixation and the uptake of external substrates. Several studies report enhanced secretion or excretion phenotypes in mutants or under specific nutrient regimes during mixotrophic growth (Zhang *et al.* 2021). The unifying principle remains consistent: when total carbon influx exceeds the cell's capacity for assimilation, storage, or balanced growth, overflow pathways become energetically favorable.

Salt stress as a potential multifactor driver of overflow: Salt stress represents a particularly complex driver of energy management because it simultaneously increases maintenance energy demand and constrains biosynthetic capacity. Ion pumping and the synthesis of compatible solutes substantially increase ATP consumption, while osmotic and ionic stress can slow growth and impair protein synthesis. Under these conditions, photosynthetic activity may persist intermittently, creating mismatches between carbon fixation and utilization that favor the export of metabolites.

Salt also reshapes carbon allocation by competing with internal sinks. Carbon is redirected toward compatible solutes such as glucosylglycerol or sucrose, which can suppress organic acid overflow in some cases. However, if osmolyte synthesis is limited by regulatory bottlenecks or temporal delays, particularly during the early stages of salt shock, fixed carbon may transiently accumulate and spill into extracellular metabolites. Additionally, salt stress can intensify photoinhibition by inhibiting the repair of PSII and increasing reactive oxygen species, further destabilizing the redox balance and promoting the need for alternative dissipation routes. System-level analyses reinforce this view. Transcriptomic and metabolomic studies reveal that salt acclimation entails a dynamic, time-dependent reorganization of metabolic networks, particularly shortly after salt exposure, encompassing pathways for the synthesis of compatible solutes and broader central metabolism. Such reorganization can generate transient overshoots in metabolite pools, thereby contributing to extracellular release.

Importantly, overflow under salt stress should not be equated with a single metabolite signature. Depending on species and growth conditions, the exometabolome may include organic acids, sugars, osmolytes, amino acids, and other small metabolites. Salt-associated overflow is therefore best viewed as a spectrum of exudation behaviors rather than a uniform response.

Sodium stress cycling represents a powerful environmental strategy to enhance carbon flux toward fermentative products in cyanobacteria. Carrieri *et al.* (2010) demonstrated that growth under high salinity induces substantial accumulation of soluble carbohydrates (up to ~50% of dry mass), while subsequent exposure to hypoionic conditions accelerates carbohydrate catabolism and significantly increases fermentation product yields, including ethanol, acetate, and formate. This is

an example under transition to low-salt conditions. Notably, hydrogen production remained unchanged, indicating a shift in redox balancing toward ethanol synthesis under osmotic stress. Mechanistically, this process is driven by osmotic regulation of metabolism, where cells redirect carbon storage and overflow pathways in response to ionic changes. These findings highlight the critical role of environmental stress in regulating metabolic overflow and carbon partitioning, offering a scalable approach to improve (Carrieri *et al.* 2010) biofuel production in cyanobacteria without genetic modification. It is important to note that the limited data to support our hypothesis and future work are needed to approve or reject the proposal on the possibility of salt stress as a potential multifactor driver of overflow in cyanobacteria.

Engineering perspective: overflow as a platform for bioproduction: From a biotechnological standpoint, overflow metabolism is increasingly framed as an opportunity rather than a liability (Fig. 3). If cyanobacteria naturally export carbon under imbalanced conditions, metabolic engineering can redirect this flux toward the production of desired products. Recent reviews on cyanobacterial organic acid production describe strategies to channel pyruvate and related branch points into lactate, acetate, acetoin, and other value-added compounds (Angermayr *et al.* 2015). In principle, salt stress could serve as a controllable lever to induce overflow-like states; however, this approach must balance enhanced export against the negative impacts of salinity on growth and photosynthetic efficiency.

Methods to measure cyanobacterial overflow (under salt stress and beyond)

Salt acclimation in the model cyanobacterium *Synechocystis* sp. PCC 6803 was analyzed by integrating transcriptomic, proteomics, and metabolic data (Carrieri *et al.* 2010). A dynamic reorganization occurred during the first hours after salt shock. The massive accumulation of glucosylglycerol then had a measurable impact on the carbon and nitrogen metabolism. The data provide the basis for engineering cyanobacterial salt tolerance and to further understand its regulation (Klähn *et al.* 2021).

At a conceptual level, metabolite overflow is defined by the accumulation of extracellular metabolites over time. In practice, however, accurately quantifying overflow in cyanobacteria, particularly under salt stress, requires careful experimental design to disentangle biological secretions from physicochemical artifacts introduced by saline matrices. As shown in Fig. 4, robust workflows therefore integrate growth and physiological measurements with targeted and untargeted exometabolomics and, when possible, complementary analyses of metabolic flux, energy status, and regulatory state.

Experimental design essentials, especially under salt stress: Appropriate controls are critical for interpreting extracellular metabolite accumulation. Media-only controls incubated under identical light and temperature conditions are crucial for capturing non-biological changes, such as salt-dependent chemical instability or evaporation-driven concentration effects. Cell-free filtrate

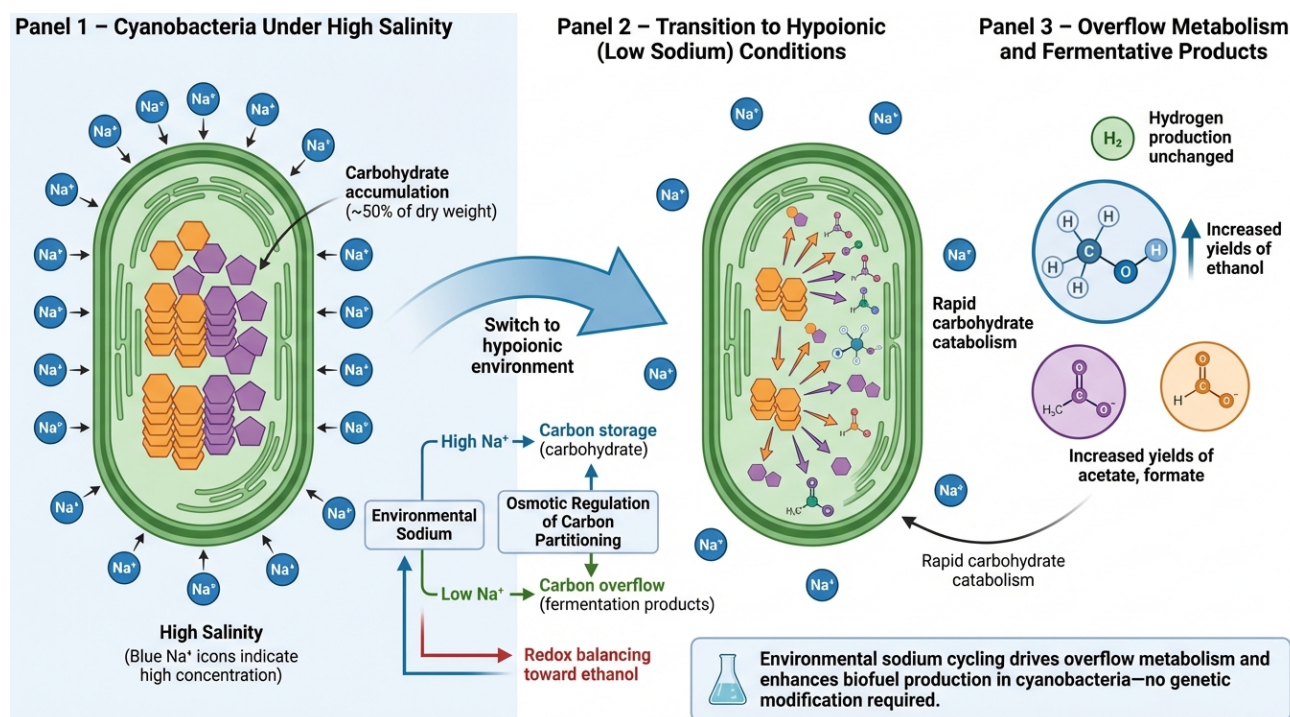


Fig. 3. Sodium stress cycling represents a powerful environmental strategy to enhance carbon flux toward fermentative products in cyanobacteria.

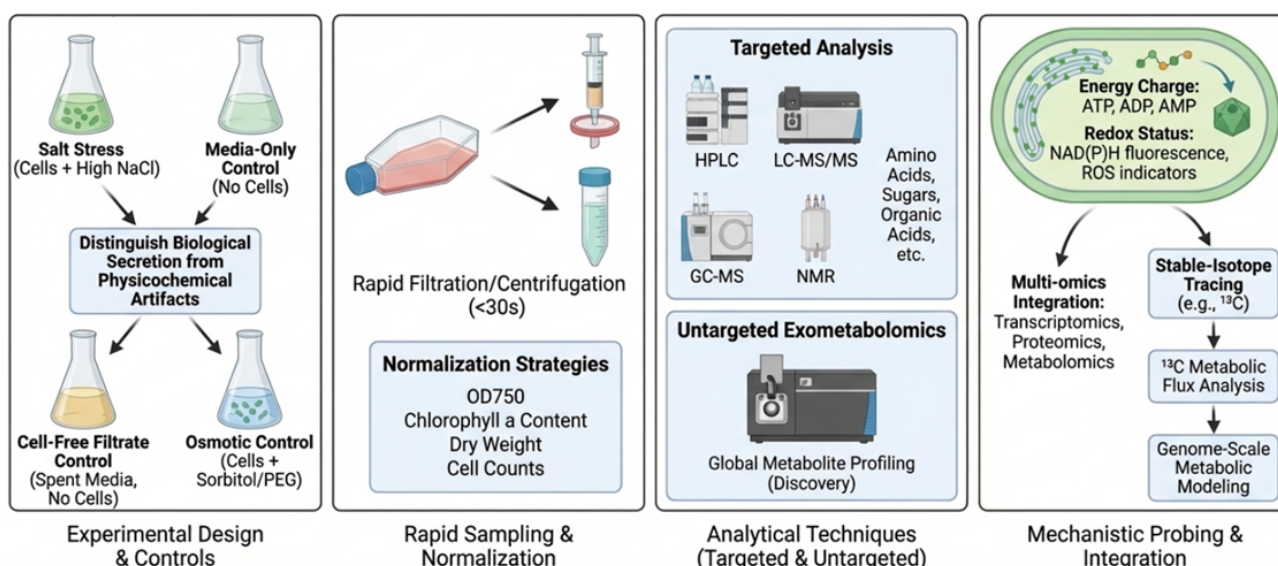


Fig. 4. Workflow of metabolite overflow in cyanobacteria under high salt conditions to reveal the response mechanisms of energy management.

controls further help verify the stability of metabolites in saline matrices after sampling. Where feasible, osmotic controls using non-ionic osmolytes can help separate osmotic from ionic effects, although their interpretation depends on the organism and the experimental question (Lu *et al.* 2017).

Sampling procedures must minimize post-harvest artifacts. Cells should be rapidly separated from the supernatant by centrifugation or filtration, and intracellular metabolism should be quenched if paired intracellular metabolomics are planned. In long-term salt experiments, evaporation should be monitored carefully, as even modest volume loss can alter effective salinity and metabolite concentrations. Normalization is equally important: extracellular metabolite accumulation is typically normalized to optical density (OD_{750}), chlorophyll content, dry mass, or cell counts. Because salt stress can alter cell morphology and aggregation, the choice of biomass proxy should be validated for each condition (Ladd *et al.* 2021). High salinity can interfere with analytical workflows, particularly liquid separation techniques, due to ionic strength and charge effects, representing a significant experimental challenge.

Targeted analytical approaches for extracellular overflow metabolites: Targeted analytical techniques remain the backbone of overflow studies. High-performance liquid chromatography (HPLC) with refractive index or UV detection is commonly used to quantify organic acids and sugars, including pyruvate, acetate, lactate, and succinate, as well as compatible solutes such as sucrose or glucosylglycerol, in some cases. HPLC is well-suited for routine time-course experiments and mutant screening due to its robustness and accessibility.

Liquid chromatography–tandem mass spectrometry (LC–MS/MS) offers higher sensitivity and selectivity and is particularly valuable when overflow metabolites are

present at low concentrations or span diverse chemical classes. LC–MS/MS enables simultaneous detection of organic acids, amino acids, osmolytes, and selected phosphorylated intermediates. Gas chromatography–mass spectrometry (GC–MS) is especially effective for volatile or derivatized compounds and is useful when expanding analyses beyond canonical overflow metabolites to include alcohols and fermentative products. Nuclear magnetic resonance (NMR) spectroscopy, while less sensitive, provides unambiguous structural identification and absolute quantification in complex saline matrices and is often used to validate findings from mass spectrometry-based methods.

Untargeted exometabolomics: Untargeted exometabolomics is increasingly employed to map the extracellular metabolite landscape under diverse environmental conditions. This approach enables the discovery of condition-specific exudate signatures and can reveal unexpected metabolites associated with stress adaptation, redox balancing, or osmoprotection (Ladd *et al.* 2021). Under salt stress, untargeted methods are particularly powerful because overflow is not restricted to central carbon intermediates but may include amino acids, osmolytes, sugars, and other stress-related metabolites. High-throughput untargeted workflows also facilitate systematic comparisons across nutrient regimes, light intensities, and salinity levels, providing a broader view of overflow phenotypes.

Probing overflow mechanisms: energy, redox, flux, and regulation: To move beyond descriptive measurements, mechanistic studies of overflow often incorporate assays of cellular energy and redox state. Measurements of ATP, ADP, and AMP allow calculation of cellular energy charge, which has been linked to overflow behavior, particularly in glycogen-deficient strains where organic

acid excretion correlates with altered energy management (Lin and Jacobs-Wagner 2022). Under salt stress, energy charge measurements help distinguish regimes dominated by carbon excess from those limited by energy availability, which may cause metabolite overflow.

Redox status can be assessed using NAD(P)H fluorescence, targeted cofactor assays, and indicators of reactive oxygen species (ROS). These measurements clarify whether overflow functions as a reductant sink that alleviates over-reduction of the photosynthetic electron transport chain. Integrating transcriptomics, proteomics, and metabolomics further strengthens mechanistic interpretation. In *Synechocystis* sp. PCC 6803, salt acclimation has been dissected using multi-omics approaches, enabling pathway-level hypotheses that link stress responses to carbon partitioning and overflow.

Stable-isotope tracing and ^{13}C metabolic flux analysis (MFA) provide quantitative insight into how carbon flux is redistributed among the Calvin cycle, glycolysis, the oxidative pentose phosphate pathway, and tricarboxylic acid (TCA) cycle branches. MFA is particularly informative when comparing low- and high-salt conditions to test whether salinity shifts flux toward overflow-prone branch points. Genome-scale metabolic models, developed for model cyanobacteria such as *Synechocystis* PCC 6803, provide a complementary framework for interpreting overflow as the outcome of constrained growth or storage capacity in the presence of sustained carbon fixation (Adebiyi *et al.* 2015).

Conclusions

Cyanobacteria advanced oxygenic photosynthesis and are valuable models to study molecular mechanisms of the photosynthetic process under different environmental conditions (Hudson 2021, Shen *et al.* 2021, Hou and Allakhverdiev 2023). During recent years, they have been increasingly considered as green cell factories for different biotechnological purposes (Hudson 2021). For example, salt stress constrains cyanobacterial growth in nature as well as in mass cultivation systems. The photosynthetic carbon fixation and cellular demand differ, leading eventually to the release of organic compounds to achieve equilibrium as an alternative energy management mechanism (Cano *et al.* 2018). Salt stress and metabolite overflow in cyanobacteria represent an underexplored intersection of physiology and metabolism. The significant contributions of Murata and Allakhverdiev *et al.* established that salt induces both osmotic and ionic stress on plant cells, which severely disrupts photosynthesis (Allakhverdiev *et al.* 2002, Marin *et al.* 2003, Allakhverdiev and Murata 2008). While existing studies suggest that carbon imbalance under stress may promote overflow-like behaviors, direct evidence under salt-acclimated steady-state conditions remains limited.

Future work integrating controlled physiological experiments with systems-level analyses will be essential to distinguish adaptive overflow from stress-induced leakage and to validate the hypotheses proposed in this review.

Future progress will depend on combining controlled physiological experiments with systems-level analyses, including multi-omics integration, stable-isotope flux measurements, and genome-scale metabolic modeling. Such approaches will be essential for disentangling cause and consequence in salt-associated overflow and for identifying regulatory nodes that can be targeted through genetic or process engineering. Future studies should also consider hypoosmotic transitions, during which rapid solute release *via* mechanosensitive channels provides a well-established mechanism for metabolite export. By framing metabolite overflow as an emergent property of carbon imbalance under stress, this review proposed a conceptual foundation for interpreting cyanobacterial behavior in saline environments and for guiding the rational design of robust strains and cultivation strategies for sustainable bioprocessing. It is important to note that direct experimental evidence linking salt acclimation to metabolite overflow under steady-state conditions remains limited, and several of the connections discussed in this review should be considered hypotheses requiring further validation.

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